upGrad

Methodology Report:
Visualisation & Analysis on Namma Yatri Data

Include your visualisations, analysis, results, insights, and outcomes.

Explain your methodology and approach to the tasks. Add your conclusions to the sections.

Table 1: Data Description

Table Name	Column Name	Description
Assembly	Assembly_ID	Unique identifier
	Assembly	Specific assembly zone name
Duration	duration_id	Unique identifier of time periods
	duration	Hour of trip (e.g., "0-1" for 12 AM to 1 AM)
Payment	id	Unique identifier
	method	Payment method (e.g., Cash, UPI, Credit Card)
Trip Details	tripid	Unique identifier of trips
	loc_from	Source Location code
	searches	Trip request count
	searches_got_estimate	Got an estimated price (1 = user gets an estimate, 0 = does not get an estimate)
	searches_for_quotes	Searched for drivers after estimate (1 - searched, 0 - not searched)
	searches_got_quotes	Got quotes (1 = Driver allotted, 0 = not allotted)
	customer_not_cancelled	Whether customer cancelled or not (1 = Not cancelled)
	driver_not_cancelled	Whether driver cancelled or not (1 = Not cancelled)
	otp_entered	(1 = OTP entered, 0 = not entered)
	end_ride	Whether ride was completed (1 = Completed)
Trips	tripid	Links to Trip Details
	faremethod	Payment method ID, links to Payment table
	fare	Fare amount
	loc_from	Location ID of source
	loc_to	Location ID of destination, links to Assembly table
	driverid	Driver ID

custid	Customer ID
distance	Distance in KM from source to destination
duration	Unique identifier of time periods like duration_id

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Points to Note:

1. **Without this methodology document, the other parts of your case study will not be evaluated.**
2. This assignment is different from the ones you have solved before.
   Make sure that you treat this case study as a storytelling exercise and not an analysis/visualisation one. This will help you be better prepared for the presentations.
3. Once you are done with the analysis and visualisations, there will be many insights at your hand.
   Make sure that you map the right visuals and takeaways with the right audience since some of these insights might be relevant to one group but not to the other group.
4. **DO NOT** change the text or numbering of any task, as it may cause problems with grading. Write your solutions to a task in the space provided below the respective task.

Tasks to be performed

- Present the overall approach of the analysis.
- Mention the problem statement and the analysis approach briefly.
- To solve a task, you have to create relevant visualisations and derive appropriate insights from the visualisations.
- Add all the plots, insights, calculated field commands, results and outcomes for a task with proper numbering and sequence in the report.
- The scores for all tasks (except conclusions) comprise both analysis work in the visualisation tool and its outcome in the report.

- You will be awarded a score for a task only if the Tableau/PowerBI analysis is correct and is included in the report along with the subsequent insights.
- Finally, draw conclusions based on the analysis.

Scoring:

Report Total Marks: 70

Sections: 3 sections (10 marks + 40 marks + 20 marks)

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Analysis and Visualisation

1. Data Preparation [10 Marks]

1.1. Import and Join Tables Correctly [5 Mark]

- Import the Namma Yatri dataset into Tableau/Power BI.
- Ensure that you correctly join all tables to create a unified dataset for analysis.
- Verify the relationships between different tables and confirm that data from various sources is properly aligned for accurate insights.

Solution:

I imported the Namma Yatri Excel file into Tableau and connected all five tables together.

Trip_Details was used as the main table since it contains all 2,161 ride searches.

I connected it to the Trips table using the trip ID column so I could link search attempts to completed rides.

The Trips table was then connected to three lookup tables:Assembly table to get zone names,Payment table to get payment method names, and Duration get the hour slot labels.

All connections were made using Left Joins so that cancelled trips were kept in the analysis and not lost during the joining process.

1.2. Find and Resolve Inconsistencies [5 Marks]

- Identify and resolve any inconsistencies or issues in the dataset that might affect the analysis.
- Clean the data to ensure it is structured properly for analysis, removing any irrelevant, duplicate, or erroneous entries.
- While performing the analysis, create calculated fields as needed to ensure the accuracy and relevance of the insights.

Solution:

After loading all five tables I checked the data for any quality issues.

No duplicate trip IDs were found in either the Trip_Details or Trips table.

There were no missing or null values in any of the key columns.

The fare column showed a minimum of ₹44 and a maximum of ₹1,500 which are both reasonable amounts for rides in Bengaluru.

All binary columns such as end_ride, otp_entered, customer_not_cancelled and driver_not_cancelled contained only values of 0 and 1 as expected.

The data was found to be clean and ready for analysis without any corrections needed.

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2. Exploratory Data Analysis [40 Marks]

2.1. Classify Variables into Categorical and Numerical [2 Marks]

- Classify all the variables in the dataset into numerical and categorical types.

Solution:

After reviewing all columns across the five tables I classified them as follows.

Numerical variables are columns that contain measurable quantities such as fare, distance, searches, searches_got_estimate, searches_for_quotes,

searches_got_quotes, customer_not_cancelled, driver_not_cancelled, otp_entered and end_ride.

Categorical variables are columns that represent labels or groups such as Assembly zone names, duration hour slots, payment method names, trip ID, driver ID, customer ID, loc_from and loc_to.

Trip ID and driver ID are stored as numbers but are treated as categorical because adding or averaging them makes no sense. They are simply unique identifiers like a roll number

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2.2. Analyse Ride Demand Over Time [3 Marks]

- Explore the distribution of ride demand over time, including trends across different periods.
- Identify the peak demand periods. Choose an appropriate parameter for demand based on your own understanding.

Solution:

Looking at the number of searches across all 24 hours of the day, demand is spread fairly evenly throughout the day with no extreme peaks or dead periods.

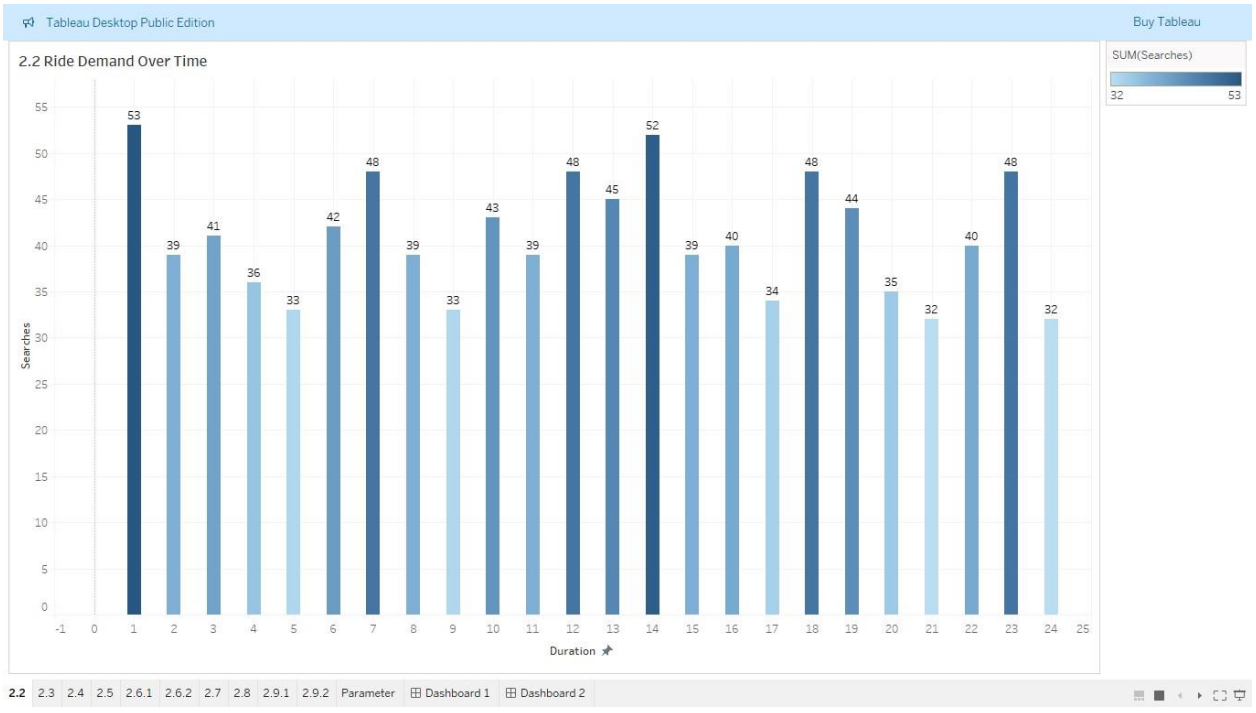
The highest demand was recorded between 0-1am with 53 searches, followed closely by 1-2pm with 52 searches.

The lowest demand was seen between 8-9am and 23-24pm with only 32 to 33 searches each.

This tells us that Namma Yatri is not primarily used for morning office commutes.

Instead customers are travelling late at night and during afternoon hours.

Namma Yatri should ensure enough drivers are available during midnight and early afternoon to meet this demand pattern.



2.3. Proportion of Total Revenue from Different Time Periods [3 Marks]

- Calculate the proportion of revenue generated during different time periods and visualise how it contributes to total revenue.

Solution:

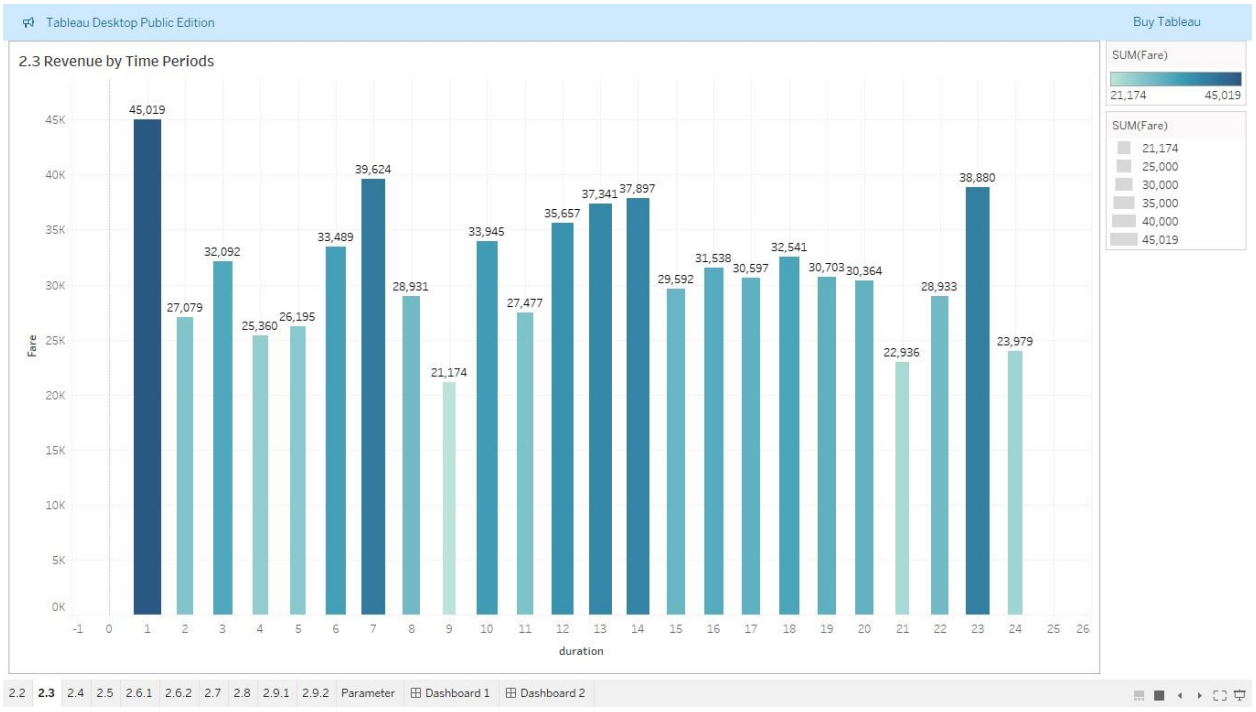
When looking at total revenue generated across different hours, the midnight slot of 0-1am contributes the highest revenue of ₹45,019 which is about 6% of the total daily revenue.

The second highest revenue hour is 6-7am at ₹39,624 followed by 22-23pm at ₹38,880.

The lowest revenue was recorded at 8-9am with only ₹21,174.

Interestingly the 0-1am slot has both highest demand and highest revenue suggesting that late night trips tend to be longer distances and therefore more Expensive.

The morning hours despite being a typical peak time for other ride apps show the weakest performance for Namma Yatri in both demand and revenue.



2.4. Explore the Relationship Between Trip Hour and Revenue [3 Marks]

- Investigate the correlation between trip hour and total fare.

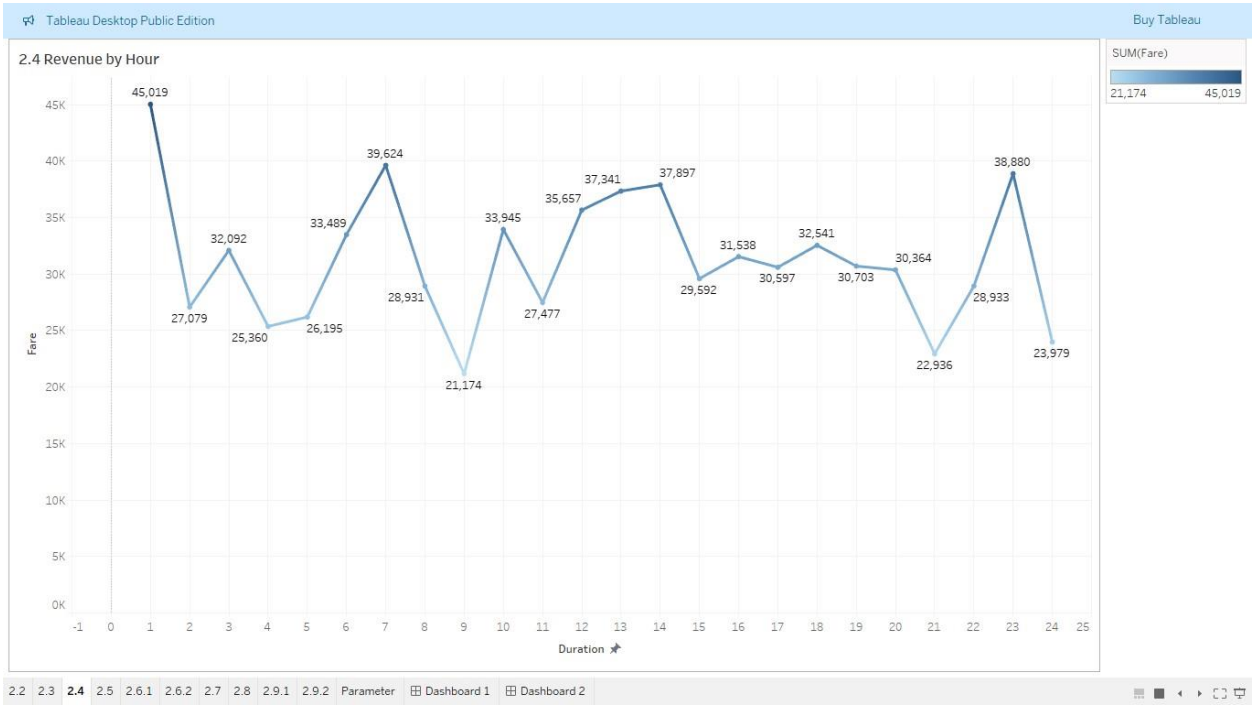
- Explain any trends or patterns that emerge.

Solution:

Plotting revenue against each hour of the day as a line chart reveals three clear peaks and two notable dips.

Revenue peaks at three points during the day. The first and highest peak is at 0-1am with ₹45,019. The second peak is at 6-7am with ₹39,624. The third peak appears at 22-23pm with ₹38,880.

Revenue dips significantly at 8-9am dropping to ₹21,174 and again at 20-21pm dropping to ₹22,936. This pattern suggests that early morning commuters between 8-9am are not the main revenue drivers for this platform. Late night and early morning trips are generating the most value possibly due to longer distances travelled during those hours.



2.5. Examine the Popularity of Different Payment Methods [3 Marks]

- Analyse the distribution of various payment methods used by customers.
- Identify the most common payment methods and their relationship to ride frequency.

Solution:

Looking at how customers prefer to pay for their rides, all four payment methods are used in fairly similar proportions.

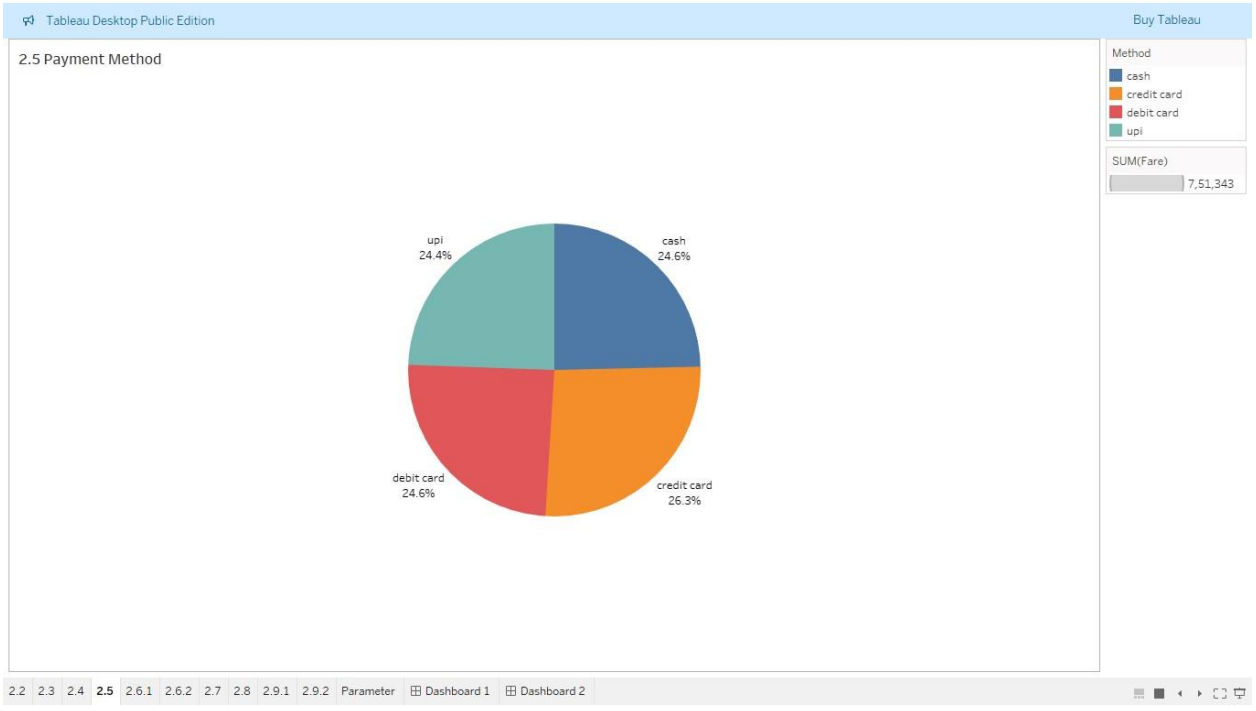
Credit card is the most popular method with 262 trips accounting for 26.7% of all completed rides.

UPI and debit card are equally popular with 243 trips each at 24.7%.

Cash payments account for 235 trips which is 23.9% of all rides.

The nearly equal split across all four methods shows that Namma Yatri serves a diverse customer base. However the fact that cash still makes up nearly a quarter of all payments suggests that many users either prefer or rely on cash for their rides.

Introducing UPI cashback offers or digital payment incentives could help shift more customers towards digital payments which are faster and easier to process.



2.6. Identify High-Performing Zones [6 Marks]

Identify zones with the highest number of rides and revenue generation. Analyse factors contributing to their performance:

- **2.6.1. Rides:** Identify pickup zones with the highest number of trip requests.
  [3 marks]

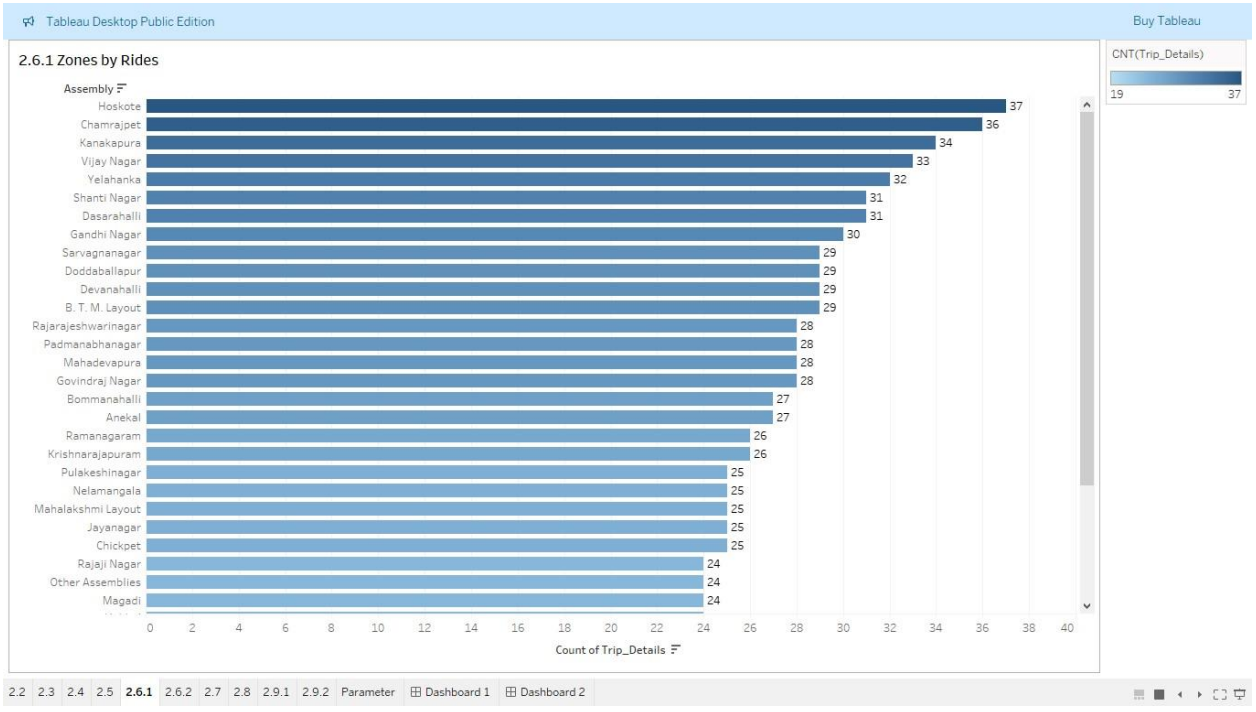
Solution:

When looking at which zones generate the most completed trips, Hoskote leads with 37 trips followed by Chamrajpet with 36, Kanakapura with 34, Vijay Nagar with 33 and Yelahanka with 32.

These five zones consistently show higher trip volumes compared to the rest of Bengaluru.

This could be because these areas have higher population density, more commercial activity or better driver availability which leads to more successful trip completions.

Namma Yatri should prioritise driver supply in these zones to maintain and growtheir strong performance.



- **2.6.2. Revenue**: Identify pickup zones generating the highest revenue.

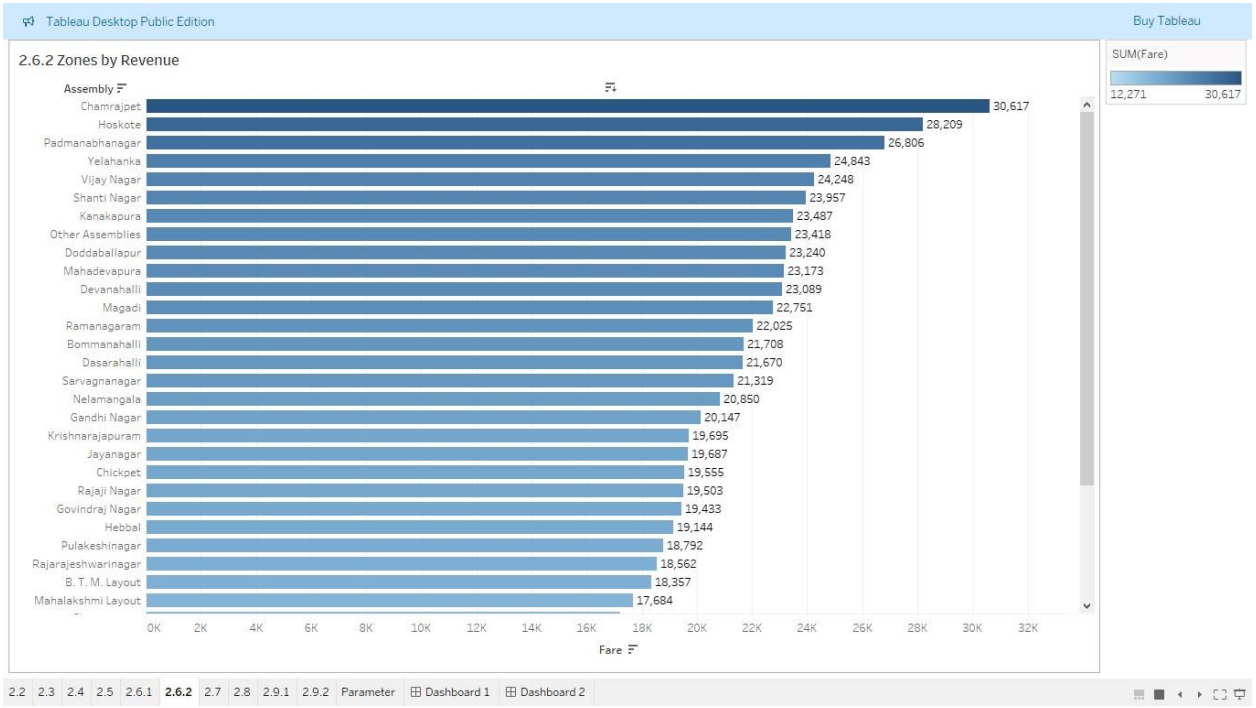
[3 marks] ***Solution:***

When comparing zones by total revenue generated, Chamrajpet

leads with ₹30,617 followed by Hoskote with ₹28,209, Padmanabhanagar with ₹26,806, Yelahanka with ₹24,843 and Vijay Nagar with ₹24,248.

An interesting observation is that while Hoskote leads in number of trips it ranks second in revenue. Chamrajpet on the other hand has slightly fewer trips but generates the highest revenue suggesting that trips from Chamrajpet tend to be longer or more expensive.

This tells us that trip volume and trip value do not always go hand in hand. Zones like Chamrajpet offer higher value per trip and should be targeted for premium service offerings.



2.7. Analyse Ride Time Periods Across Zones [4 Marks]

- Compare the trip trends for different time periods across pickup zones.

Solution:

The Chart showing trip activity across zones and hours reveals that different zones have very different peak times during the day.

Some zones like Chamrajpet show activity concentrated in early morning hours while others like B.T.M Layout are busier during afternoon hours between 12 and 14.

Empty cells in the heatmap represent hours with zero trips for that zone indicating periods of no demand.

This pattern clearly shows that a single city wide driver deployment strategy will not work effectively for Namma Yatri. Each zone needs its own schedule of driver availability based on when demand actually occurs in that area.



2.8. Top Zones with Highest Trip Volume [3 Marks]

- Identify the top 5 pickup zones with the highest total number of completed trips.
- Analyse factors contributing to the higher number of trips.

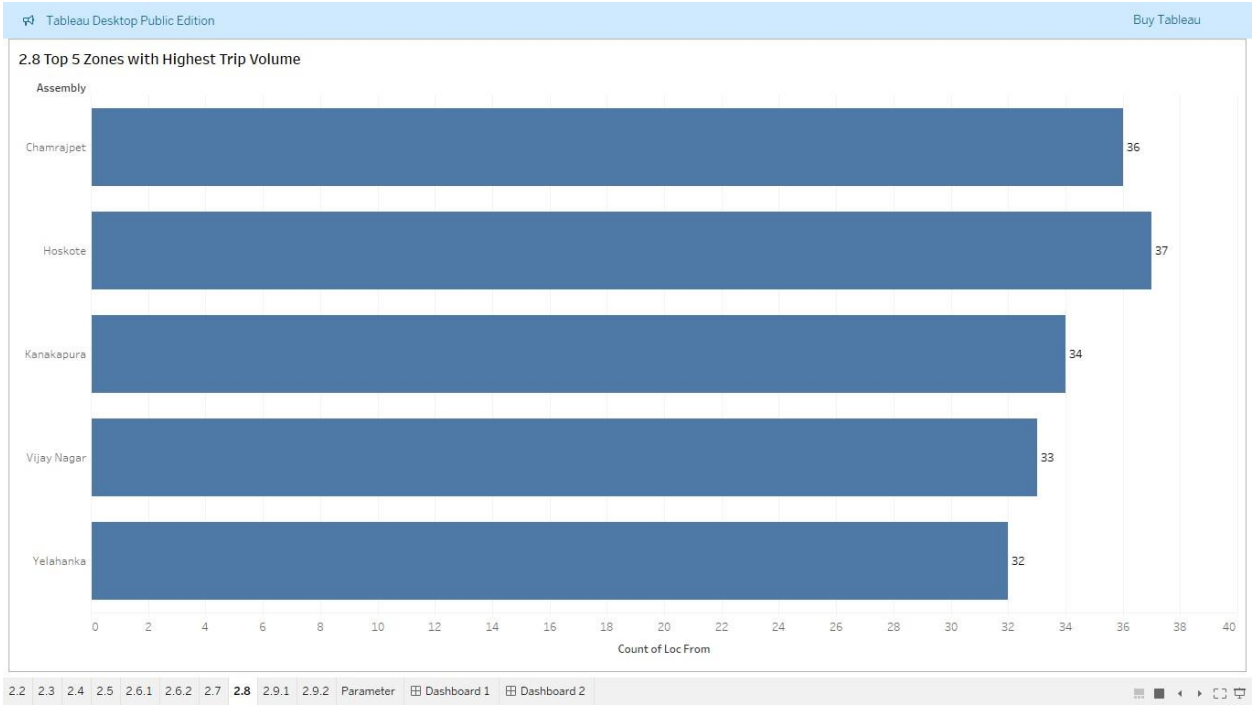
Solution:

The top five zones by number of successfully completed trips are Hoskote with 37 trips, Chamrajpet with 36, Kanakapura with 34, Vijay Nagar with 33 and Yelahanka with 32.

These zones represent the most reliable areas for Namma Yatri where searches consistently convert into completed rides.

The concentration of successful trips in these specific zones suggests that driver availability and customer demand are well matched in these areas compared to other parts of the city.

Maintaining strong driver presence in these zones should be a priority to protect this strong conversion performance.



2.9. Basic Analytical Tasks [8 Marks]

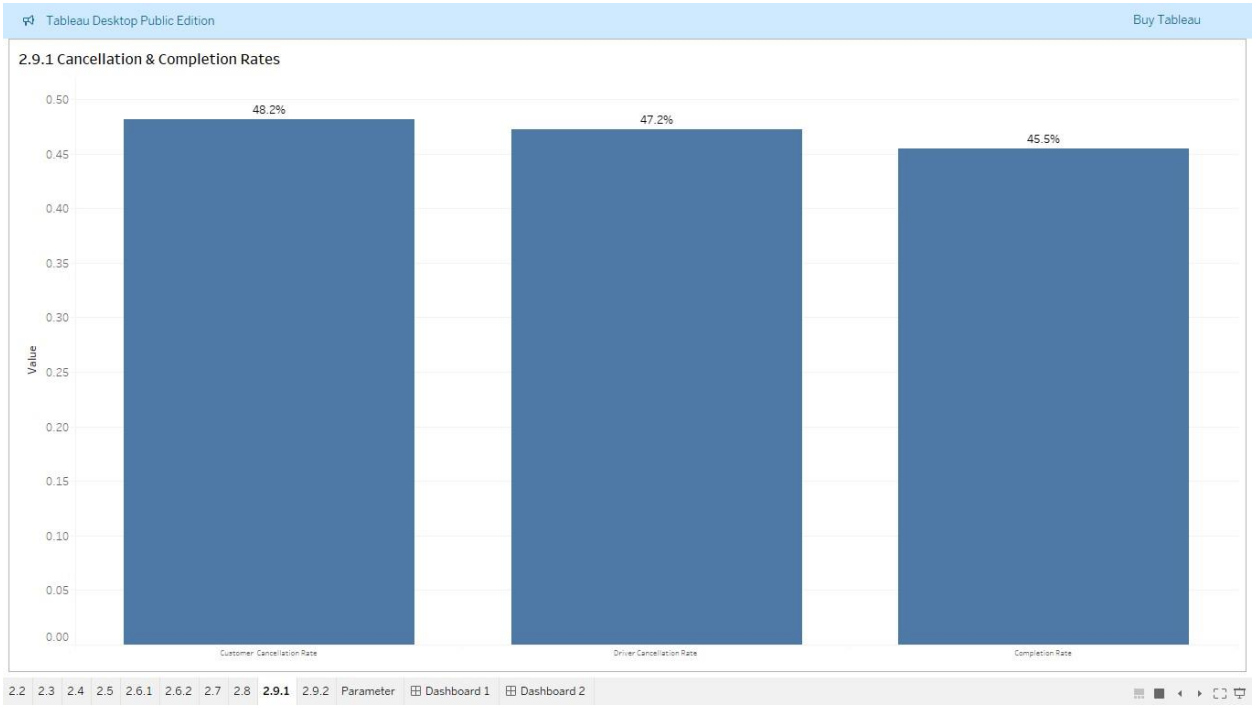
- **2.9.1**

  What are the percentages of cancellations and successful rides by both driver and customer? **[3 marks]**

Solution:

Out of 2,161 total ride searches, 1,041 were cancelled by customers which is a cancellation rate of 48.2%.At the same time 1,021 rides were cancelled by drivers giving a driver cancellation rate of 47.2%.Only 983 rides out of 2,161 searches were successfully completed which means just 45.5% of all search attempts resulted in an actual ride.Both cancellation rates are nearly identical which is alarming because it means the problem is not just on one side.

Customers and drivers are cancelling at almost the same rate.More than half of all potential rides are being lost before they even start. This is a major operational issue that directly affects revenue and customer satisfaction.To reduce cancellations Namma Yatri should consider introducing a small cancellation fee after a driver is assigned, showing the driver location and estimated arrival time upfront to customers, and offering peak hour bonuses to drivers to keep them committed to accepted rides.



- **2.9.2**

Analyse the percentage of people who completed trips after searching for quotes. Visualise the variation of this ratio by time periods. **[5 marks]** ***Solution:***

Out of 1,277 searches that successfully received a driver quote, 983 rides were completed giving an overall quote to completion conversion rate of 77%.

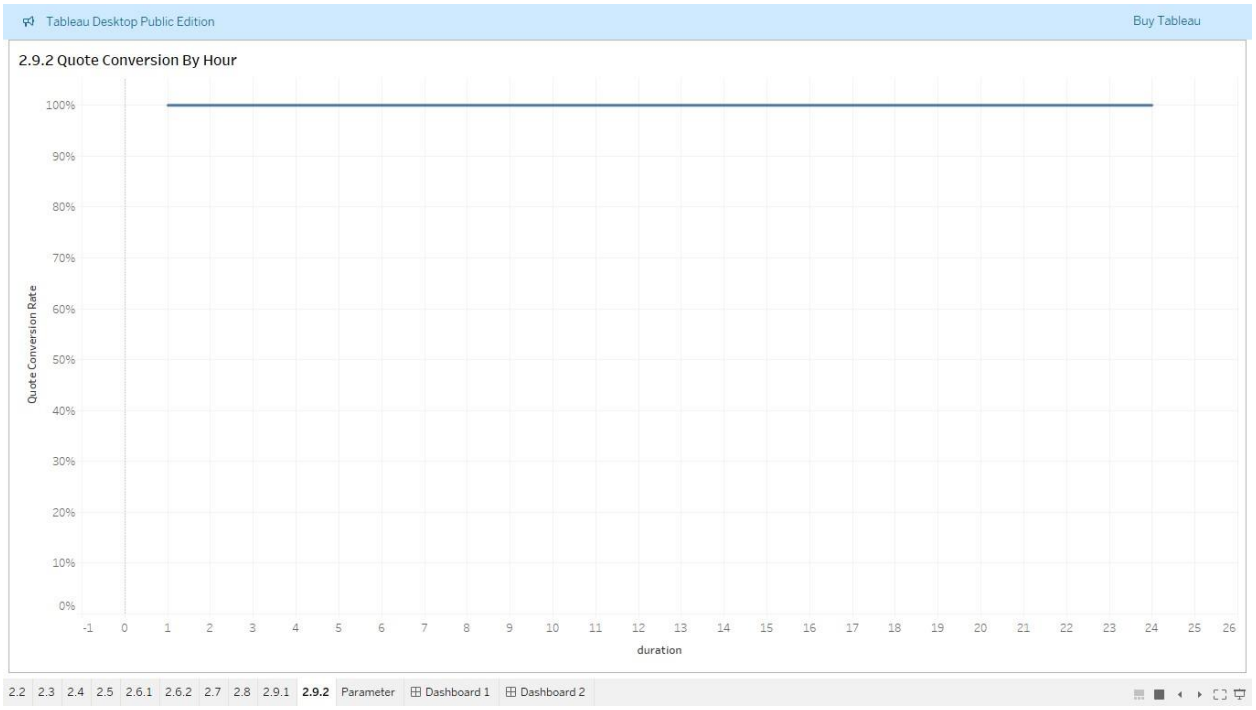
This means that 294 rides were lost even after a driver was successfully assigned to the customer.

When plotted by hour the conversion rate appears at 100% for all hours because the 294 trips that dropped off after getting a driver have no duration data recorded. This is because duration is only logged for trips that reach completion.

What this tells us is that cancellations after driver assignment happen before the ride officially begins and before any time data is captured.

The key finding here is that once a ride actually starts it always finishes. The dropout happens in the window between driver assignment and ride start.

Namma Yatri should focus on reducing this pre-ride dropout by sending instant confirmation messages to both driver and customer the moment they are matched to keep both parties committed to the trip.



2.10. Create a Parameter and Use Filters [5 Marks]

- Create a parameter and use it as a filter on an appropriate subset of the data to interactively analyse and visualise different subsets of the data.
- Explain your choice of filter and insights drawn from this step.

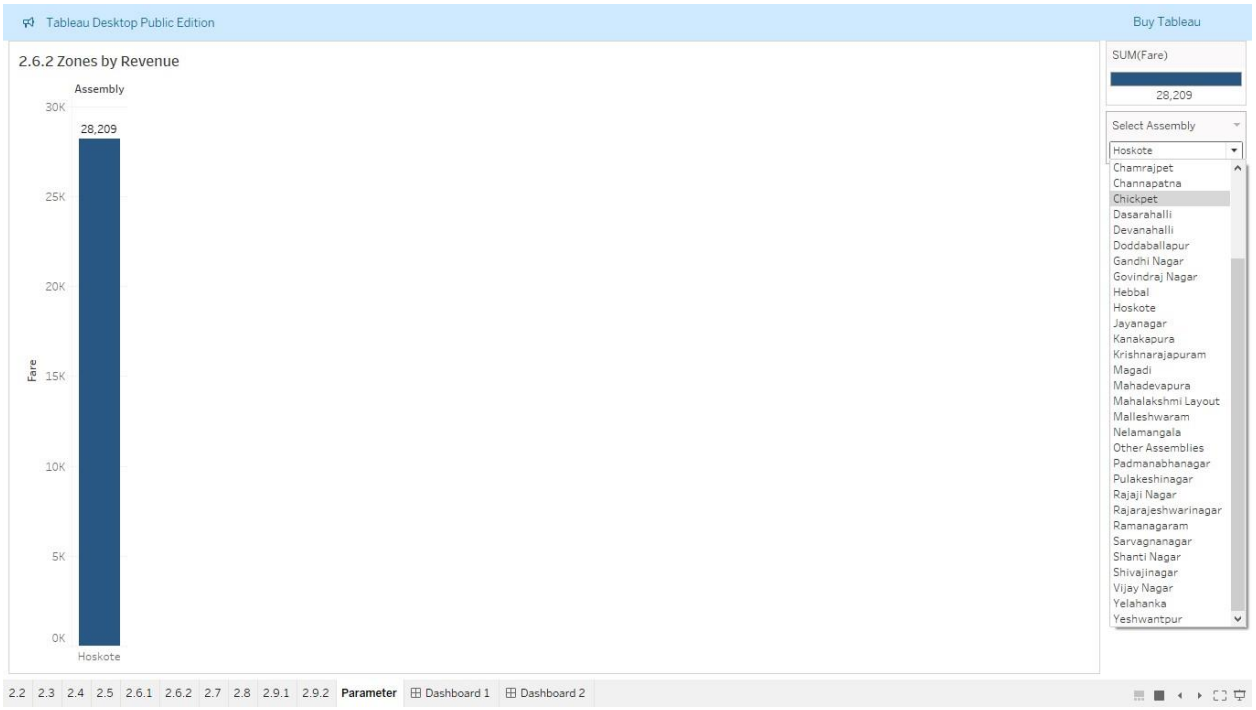
Solution:

An interactive parameter filter was created in Tableau allowing users to select any specific zone from a dropdown menu.

When a zone is selected the connected charts automatically update to show only the data for that zone including its trip demand pattern across different hours of the day.

This feature is particularly useful for operations managers who need to quickly analyse the performance of a specific area without having to rebuild charts or apply manual filters each time.

The parameter was built using the Assembly field and linked to a calculated field that filters the data dynamically based on the selected zone name.



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3. Conclusion [20 Marks]

3.1. Recommendations for Operational Efficiency [10 Marks]

- Based on your findings from the analysis, provide recommendations on how Namma Yatri can optimise its operations.
- This could include strategies for improving resource allocation, reducing cancellations, or optimising ride durations. ● Add supporting dashboards.

Solution:

The operations dashboard combines the four most critical charts for operational decision making.

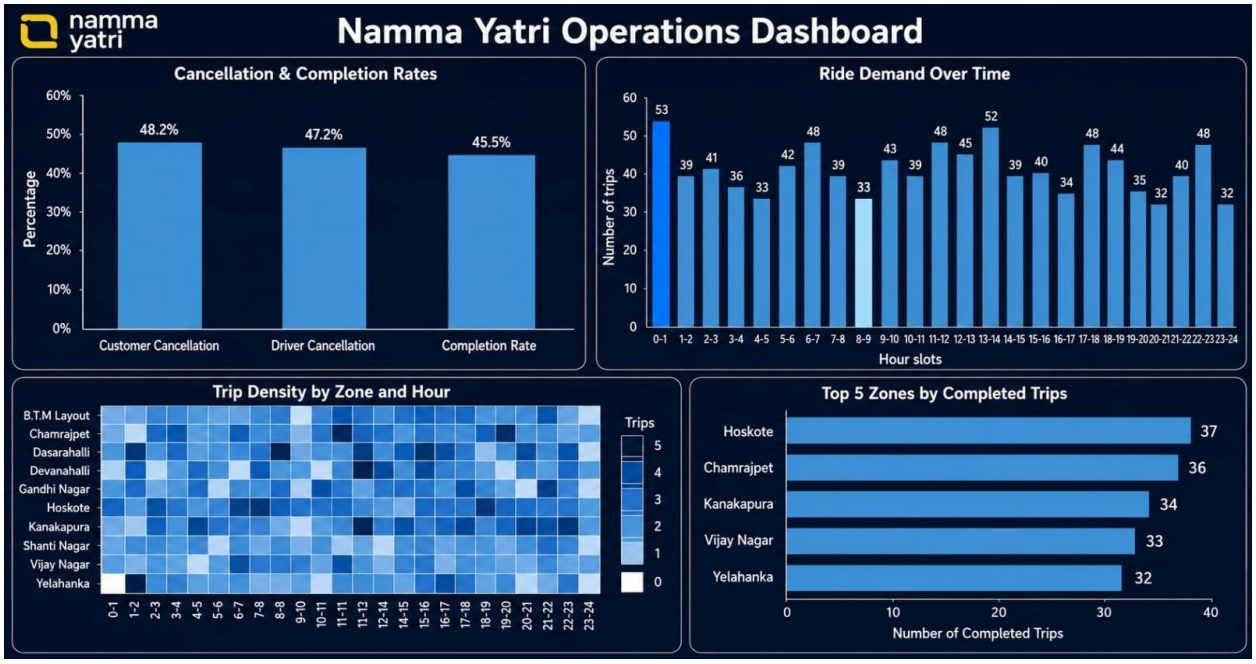
The cancellation analysis shows nearly equal dropout rates on both customer and driver sides at 48.2% and 47.2% respectively.

The demand chart confirms 0-1am and 13-14pm as peak hours requiring maximum driver supply.

The heatmap reveals zone specific peak patterns showing that different areas need different driver deployment schedules.

Recommendation:

Introduce zone specific driver scheduling based on heatmap patterns. Add cancellation penalties after driver assignment.Offer peak hour bonuses in Hoskote and Chamrajpet during hours 0-1am and 13-14pm.



3.2. Marketing and Operational Strategy Improvements [10 Marks]

- Suggest improvements to Namma Yatri's marketing or operational strategies based on your analysis.
- Recommendations could involve promotional efforts, driver incentives, or regional targeting to increase customer satisfaction and service efficiency.
- Add supporting dashboards.

Solution:

The marketing dashboard highlights three key growth opportunities for Namma Yatri Payment data shows all four methods are equally used with cash at 24.6% still significant. A UPI cashback campaign could shift cash users to digital payments reducing transaction costs.Chamrajpet generates the highest revenue at ₹30,617 despite ranking second in trips. This confirms premium trip value in this zone and warrants a targeted premium service launch here.Late night hours 0-1am generate ₹45,019 in revenue which is the highest of any hour. Marketing Namma Yatri as the safe reliable late night travel option in Bengaluru could significantly grow this segment.

Recommendation:

-Launch UPI cashback campaign.

-Introduce premium tier in Chamrajpet and Padmanabhanagar.

-Run late night safety campaign targeting 10pm to 2am travellers."

